

William Van Eck

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EDUCATION

The Pennsylvania State University

College of Engineering

Bachelor of Science in Computational Data Science

University Park, PA

May 2026

PROJECT EXPERIENCE

Poker Prediction Market

Software Engineer

University Park, PA

January 2026 – May 2026

- Created an interactive prediction market platform where users place probability-based bets on outcomes in simulated poker games generated through Monte Carlo bot gameplay.
- Designed dynamic contract pricing logic that updates market odds in real time based on changing probabilities of in game outcomes.
- Integrated an AI assistant that provides users with explanations, strategy, and recommendations based on the game state and scenarios available.

VisionScorer

Computer Vision Engineer

University Park, PA

January 2025 – May 2025

- Developed a YOLOv11 based computer vision system for basketball object detection, training models on two independently curated datasets containing 5,000 and 170 labeled images.
- Preprocessed image datasets through resizing, normalization, and augmentation to improve model consistency and reduce overfitting.
- Evaluated the impact of dataset size versus annotation quality by comparing model performance across noisy large-scale labels against smaller high quality labels using mAP performance as a metric, achieving scores of 89.9% and 75.1% respectively.
- Conducted hyperparameter tuning to optimize performance across datasets with varying sizes and label quality.

Flight Delay Prediction

Data Wrangler

University Park, PA

November 2025 – December 2025

- Collected and processed millions of flight records from 2015-2024 from the Bureau of Transportation Statistics, focusing on 5 airports around the State College area.
- Engineered predictive features including holiday indicators to prepare the flight dataset for machine learning models predicting delays and cancellations.
- Built and evaluated Random Forest and XGBoost classification models achieving F1 scores of .71 and .69 respectively for flight delay prediction.

WORK EXPERIENCE

Amazon

Amazon Associate

Bellefonte, PA

February 2026 - Present

- Processed up to 13,000 packages per day maintaining speed, accuracy, and company standards in a fulfillment environment.

UPS

Driver/Loader

Harrisburg, PA

June 2026 – August 2026

- Loaded and organized delivery trucks to support efficient routing, timely package distribution, and accuracy.

SKILLS

Programming: Python (Pandas, Scikit-learn, NumPy), R, SQL, Scala

Tools & Platforms: Git, Jupyter, Spark, Hadoop, HDFS, YOLO

Techniques: Machine Learning, Computer Vision, Feature Engineering, Classification, Data Preprocessing